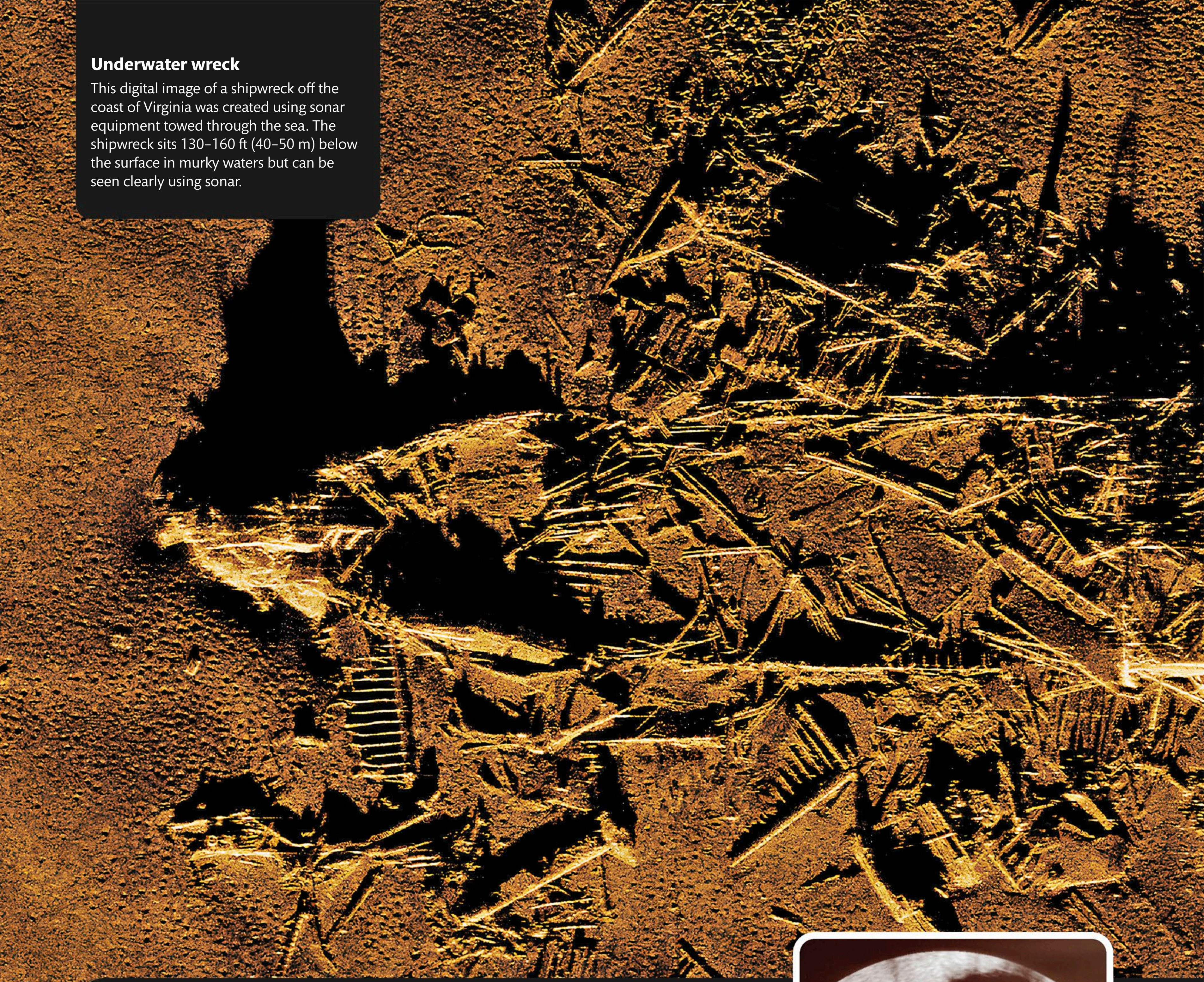

Underwater wreck

This digital image of a shipwreck off the coast of Virginia was created using sonar equipment towed through the sea. The shipwreck sits 130–160 ft (40–50 m) below the surface in murky waters but can be seen clearly using sonar.

SEEING WITH SOUND

SONAR

Sonar (Sound Navigation and Ranging) is a technology that uses sound waves to sense objects. A device called a transducer emits pulses of sound waves and another called a detector listens for their echoes, which are used to measure the distance to the object. Using this information, a computer can also make a model of the object. Sonar is often used underwater—where sound can travel great distances—for mapping the seafloor and avoiding obstacles.

SONOGRAM

Very high-pitched sound waves (too high for humans to hear) are used in healthcare, in a technique similar to underwater sonar. Sound waves are reflected off different parts in the human body to build an image called a sonogram. These images can be used to check the development of unborn babies.